

## ORIGINAL ARTICLE

# Marburg Virus Disease in Rwanda, 2024 — Public Health and Clinical Responses

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## ABSTRACT

**BACKGROUND**

On September 27, 2024, Rwanda reported an outbreak of Marburg virus disease (MVD), after a cluster of cases of viral hemorrhagic fever was detected at two urban hospitals.

**METHODS**

We report key aspects of the epidemiology, clinical manifestations, and treatment of MVD during this outbreak, as well as the overall response to the outbreak. We performed a retrospective epidemiologic and clinical analysis of data compiled across all pillars of the outbreak response and a case-series analysis to characterize clinical features, disease progression, and outcomes among patients who received supportive care and investigational therapeutic agents.

**RESULTS**

Among the 6340 patients with suspected MVD who underwent testing, 66 had laboratory-confirmed MVD, 51 (77%) of whom were health care workers. The median estimated incubation period was 10 days (interquartile range, 8 to 13), and symptom onset occurred a median of 2 days (interquartile range, 1 to 3) before hospital admission. The results of epidemiologic investigations were highly suggestive of a zoonotic origin of the outbreak: an index patient was identified who had been exposed to Egyptian fruit bats at a mining site. The case fatality rate in the outbreak was 23% (15 deaths among 66 patients). Remdesivir and the monoclonal antibody MBP091 were used under expanded access and clinical trial protocols. In addition, 1710 frontline workers and high-risk contacts received the chimpanzee adenovirus 3–vectored vaccine ChAd3-MARV under emergency use authorization in a phase 2 clinical trial.

**CONCLUSIONS**

Implementation of containment measures, advanced supportive care, and access to investigational countermeasures may have contributed to reduced mortality from MVD in this outbreak. Enhancing surveillance, improving infection prevention and control in health care settings, and ensuring timely deployment of medical countermeasures will be critical for mitigating the effects of future filovirus disease outbreaks.

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\*The members of the Rwanda MVD Outbreak Response Team are listed in the Supplementary Appendix, available at [NEJM.org](https://www.nejm.org).

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**T**HE FIRST OUTBREAK OF MARBURG VIRUS disease (MVD) reported in Rwanda was declared on September 27, 2024.<sup>1</sup> The outbreak was identified after a cluster of cases of severe febrile illness occurred in the capital city of Kigali, home to 1.7 million inhabitants. Within 1 week, MVD had affected more than 20 health care workers at two tertiary care hospitals in Kigali. Government agencies and their partners promptly implemented comprehensive response measures, including establishment of a national MVD treatment unit, introduction of investigational therapies and vaccines, enhancement of surveillance, and implementation of contact tracing and public awareness campaigns.<sup>2</sup>

Marburg virus, a member of the family *Filoviridae*, which also includes the Ebola virus, was first discovered during an outbreak in 1967 that affected laboratory personnel in two European countries.<sup>3,4</sup> MVD is a zoonotic infection transmitted to humans from fruit bats (*Rousettus aegyptiacus*), the natural reservoir.<sup>5</sup> Human-to-human transmission can occur through direct contact with bodily fluids of infected persons.<sup>6</sup> MVD has an incubation period that ranges from 2 to 21 days, although in most reported cases symptom onset occurs between 5 and 10 days after exposure.<sup>7</sup> The clinical presentation is variable and can include fever, gastrointestinal symptoms, coagulopathy, and multiorgan failure.<sup>8</sup>

MVD outbreaks have historically originated from spillover events in remote areas before potentially spreading to populated regions.<sup>9,10</sup> Large outbreaks have occurred in the Democratic Republic of Congo from 1998 through 2000 (154 cases; case fatality rate, 83%),<sup>11</sup> and in Angola in 2004 and 2005 (252 cases; case fatality rate, 90%).<sup>9</sup> Multiple outbreaks occurred in Uganda between 2007 and 2017.<sup>12</sup>

Unlike previous MVD outbreaks, which occurred in remote settings, the 2024 Rwanda outbreak emerged in two large hospitals in the capital city of Kigali. This setting presented distinct challenges, including nosocomial transmission and strain on the health care workforce, but also enabled the deployment of advanced clinical interventions. In this report, we describe the epidemiology, clinical presentations, treatment strategies, and overall public health response to the outbreak.

## METHODS

### STUDY DESIGN AND DATA SOURCES

To provide a comprehensive account of the outbreak, we conducted an integrated analysis using both prospectively and retrospectively collected data. Data sources included epidemiologic surveillance, clinical case, and laboratory records. We characterized the transmission dynamics of the outbreak over time and conducted a complete clinical case review involving all patients with laboratory-confirmed MVD.

Testing and treatment data were captured in the District Health Information System 2 (DHIS2), the comprehensive public health database maintained by the Ministry of Health for collecting routine case-level data. Clinical data were documented in the electronic medical records of hospitals in which patients with MVD received care or in paper-based bedside charts at the national treatment center that had been set up in response to the outbreak. From these sources, a team of physicians systematically extracted data elements for the case-series analysis; the data were linked by patient identifiers with MVD case records in the DHIS2.

### LABORATORY TESTING

Specimens from patients with suspected MVD were tested for viral RNA at the Rwanda National Reference Laboratory (laboratory assays are described in the Supplementary Appendix, available with the full text of this article at NEJM.org). Whole-genome sequencing was performed on Marburg virus–positive specimens to characterize the outbreak lineage, assess genetic variation, and infer transmission dynamics, as described elsewhere.<sup>13</sup>

### STATISTICAL ANALYSIS

To ensure data accuracy and completeness, we implemented a standardized approach to case documentation, integrating manual abstraction from patient charts with automated data extraction from electronic health systems. Exploratory and descriptive analyses were conducted with R software, version 4.4.2 (R Foundation for Statistical Computing), and Python, version 3.12.8 (Python Software Foundation). The Rwanda National Ethics Committee provided the approval for

this work. The authors vouch for the accuracy and completeness of the data in this report.

## RESULTS

### OUTBREAK DETECTION AND EARLY RESPONSE

On September 25, 2024, clinicians at an academic tertiary care hospital in Kigali (Hospital 1) notified officials at the Rwanda Biomedical Center of a cluster of cases of severe febrile illness among patients admitted to the intensive care unit (ICU). The patients presented with rapidly progressive fever, headache, and gastrointestinal symptoms, followed by multiple-organ failure and death within 48 hours after admission. Within 2 days, similar symptoms developed in numerous health care workers at Hospital 1. Several clinical and postmortem specimens were tested for viral hemorrhagic fevers at the National Reference Laboratory. On September 27, positive results of reverse-transcriptase–polymerase-chain-reaction (RT-PCR) testing for Marburg virus were confirmed and a coordinated national emergency response plan was activated. A national treatment center was established on the same day in a temporary mobile hospital near Kigali. On October 5, patients were transferred to a permanent facility, which included ICU capabilities and separate isolation areas.

RT-PCR testing for Marburg virus was recommended for all patients presenting for care at any health facility in Rwanda whose illness met the standardized definition of a suspected case (Table S1 in the Supplementary Appendix). Protocols for symptom screening, triage, and isolation of patients with suspected MVD were shared with all health facilities. Rapid response teams were deployed to assist local staff with key response actions, in accordance with World Health Organization (WHO) guidance.<sup>14</sup> Dedicated medical transport teams were established to transport patients to an isolation facility while their test results were pending. Contact-tracing efforts rapidly expanded, and 1497 contacts were identified and monitored, including family members, hospital staff, and other patients who had potential exposure.

The Rwandan community health worker program<sup>15</sup> conducted systematic weekly screenings across all 30 districts, reaching 1,288,205 house-

holds and 4,793,212 persons (37% of the population) to provide information on MVD. Community health workers identified 24 persons with suspected MVD who were on the contact list, all of whom tested negative by RT-PCR. Laboratory testing capacity was expanded on October 1 to include RT-PCR testing at five regional laboratory hubs, which were able to provide test results with a turnaround time of less than 12 hours. A national mortality surveillance system was established, which mandated MVD testing for all persons who died in the community during the outbreak. This effort resulted in screening of 240 decedents, all of whom tested negative, a finding consistent with transmission having remained within identified chains.

Figure 1 shows the epidemic curve for the outbreak, with the date of symptom onset for each patient with laboratory-confirmed MVD and the major milestones in the public health response. From September 25 until November 8, a total of 6340 persons were tested for Marburg virus, 66 of whom were confirmed to be positive, yielding an overall test positivity percentage of 1% (Table 1). The first case was confirmed on September 27. The majority of cases (76%) and deaths (80%) occurred within the first 2 weeks after the outbreak was reported (Fig. S1). Of the 66 patients with confirmed MVD, 51 (77%) were health care workers.

### PUBLIC HEALTH INVESTIGATIONS

To determine the source of the outbreak, a team of clinicians and epidemiologists conducted a retrospective chart review of 120 patients admitted to the ICU at Hospital 1 between July 1 and September 25. This review began in the second week of the outbreak, after initial investigations of the first two identified cases (in Patients C000 and C001) showed no evidence of direct zoonotic exposure or previous contact with symptomatic persons. Investigators systematically reviewed records of ICU patients with a body temperature greater than 38°C and at least one additional feature suggestive of viral hemorrhagic fever.

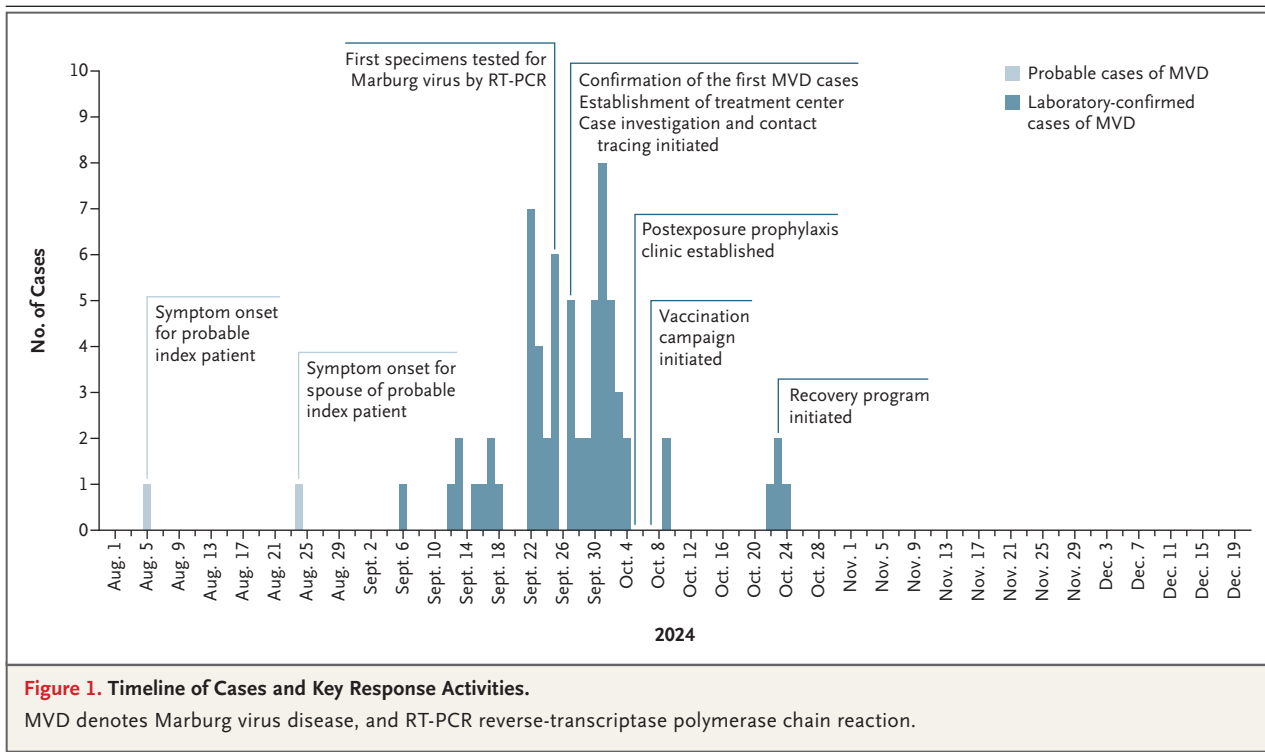
The retrospective review identified one case that was highly suspicious for undiagnosed MVD, in a 25-year-old woman who had been transferred to Hospital 1 on August 26 after undergoing a cesarean delivery at a district hospital.

Fever, bleeding, elevated aminotransferase levels, and thrombocytopenia had developed after delivery, prompting transfer to Hospital 1 with presumptive diagnosis of HELLP syndrome (characterized by hemolysis, elevated liver enzyme levels, and low platelet counts). On arrival, she had acute kidney injury, coagulopathy, and respiratory distress. Despite resuscitation efforts, she died within 24 hours after ICU admission, on August 27. Viral hemorrhagic fever was not considered as a diagnosis at the time of her death, and no specimens were retained for subsequent testing. Admission records showed that the first patient with confirmed MVD (Patient C000), a 37-year-old man hospitalized for diabetic ketoacidosis, was admitted to the ICU bed adjacent to hers shortly after her death.

The epidemiologic investigation linked the deceased woman to her husband, a 27-year-old miner working in a rural mining site with a tunnel inhabited by Egyptian fruit bats. The miner described having had an illness with fever, headache, and fatigue in early August 2024, but he had recovered without seeking medical care. He reported that his wife began to have similar symptoms in late August, shortly after delivery, and their infant also had died from an unexplained

febrile illness 1 week after birth. Serologic testing of a blood specimen from the husband revealed a high titer of Marburg virus–specific IgG antibody (1:6400) with no detectable IgM, which suggested past infection. Investigators concluded that he was likely to be the index patient, with infection resulting from zoonotic spillover and subsequent transmission to his wife and newborn. Patient C000 was probably exposed to the virus through contaminated surfaces while he was receiving care in the ICU of Hospital 1.

Integration of contact-tracing data with field investigations of probable preoutbreak cases enabled a comprehensive characterization of transmission networks during the outbreak (Fig. 2). Three prominent transmission clusters were identified. The first cluster involved patients and staff at Hospital 1 who were most likely exposed to Patient C000 or to Patient C001, a health care aide who provided direct care to Patient C000 during his illness. The second cluster emerged at Hospital 2, where Patient C002 — a physician involved in resuscitation attempts for Patient C000 — later worked in patient-care areas, facilitating additional transmission. The third cluster primarily consisted of family members and other close contacts of Patient C000, many of



**Table 1. Demographic Characteristics and Test Results in Patients with Suspected Marburg Virus Disease (MVD).**

| Characteristic                 | Patients with<br>Suspected MVD<br>(N = 6340) | Test Result                               |              |
|--------------------------------|----------------------------------------------|-------------------------------------------|--------------|
|                                |                                              | Negative                                  | Positive     |
|                                |                                              | number of patients/total number (percent) |              |
| Age                            |                                              |                                           |              |
| <10 yr                         | 386/6159 (6)                                 | 385/386 (>99)                             | 1/386 (<1)   |
| 10–19 yr                       | 776/6159 (13)                                | 775/776 (>99)                             | 1/776 (<1)   |
| 20–29 yr                       | 1890/6159 (31)                               | 1870/1890 (99)                            | 20/1890 (1)  |
| 30–39 yr                       | 1444/6159 (23)                               | 1414/1444 (98)                            | 30/1444 (2)  |
| 40–49 yr                       | 828/6159 (13)                                | 817/828 (99)                              | 11/828 (1)   |
| ≥50 yr                         | 835/6159 (14)                                | 832/835 (>99)                             | 3/835 (<1)   |
| Sex                            |                                              |                                           |              |
| Female                         | 2931/6235 (47)                               | 2910/2931 (99)                            | 21/2931 (1)  |
| Male                           | 3304/6235 (53)                               | 3259/3304 (99)                            | 45/3304 (1)  |
| Occupation                     |                                              |                                           |              |
| Health care worker             | 246/3480 (7)                                 | 196/246 (80)                              | 51/246 (20)  |
| Non–health care worker         | 3234/3480 (93)                               | 3218/3234 (>99)                           | 15/3234 (<1) |
| Province of residence          |                                              |                                           |              |
| City of Kigali                 | 4018/6168 (65)                               | 3960/4018 (99)                            | 58/4018 (1)  |
| Eastern Province               | 657/6168 (11)                                | 656/657 (>99)                             | 1/657 (<1)   |
| Northern Province              | 545/6168 (9)                                 | 545/545 (100)                             | 0            |
| Southern Province              | 533/6168 (9)                                 | 527/533 (99)                              | 6/533 (1)    |
| Western Province               | 415/6168 (7)                                 | 414/415 (>99)                             | 1/415 (<1)   |
| Week of outbreak               |                                              |                                           |              |
| Week 1, September 23–29        | 157/6338 (2)                                 | 130/157 (83)                              | 27/157 (17)  |
| Week 2, September 30–October 6 | 1401/6338 (22)                               | 1376/1401 (98)                            | 25/1401 (2)  |
| Week 3, October 7–13           | 988/6338 (16)                                | 979/988 (99)                              | 9/988 (1)    |
| Week 4, October 14–21          | 949/6338 (15)                                | 948/949 (>99)                             | 1/949 (<1)   |
| Week 5, October 22–29          | 709/6338 (11)                                | 705/709 (99)                              | 4/709 (1)    |
| Week 6, October 30–November 6  | 1383/6338 (22)                               | 1383/1383 (100)                           | 0            |
| Week 7, November 7–14          | 604/6338 (10)                                | 604/604 (100)                             | 0            |
| Week 8, November 15–22         | 147/6338 (2)                                 | 147/147 (100)                             | 0            |

whom visited him in the hospital before the outbreak was recognized.

A single case identified on October 24 (in Patient C064) was not linked to any of the three clusters. The infection of this patient, a miner who worked in the same tunnel as the index patient, was suspected to represent a second direct zoonotic spillover event, but confirmation by viral sequencing was precluded by a high RT-PCR cycle threshold (Ct) (i.e., a low viral load). Sero-

logic testing of 137 miners who worked in the same tunnel identified 14 additional persons with detectable Marburg virus–specific IgG antibodies, which suggested the occurrence of unrecognized spillover events and potential subclinical infections within the mining community.

#### CLINICAL MANAGEMENT OF CONFIRMED MVD

Among the 66 patients with confirmed MVD, the median age was 31.5 years, and 21 (32%) were

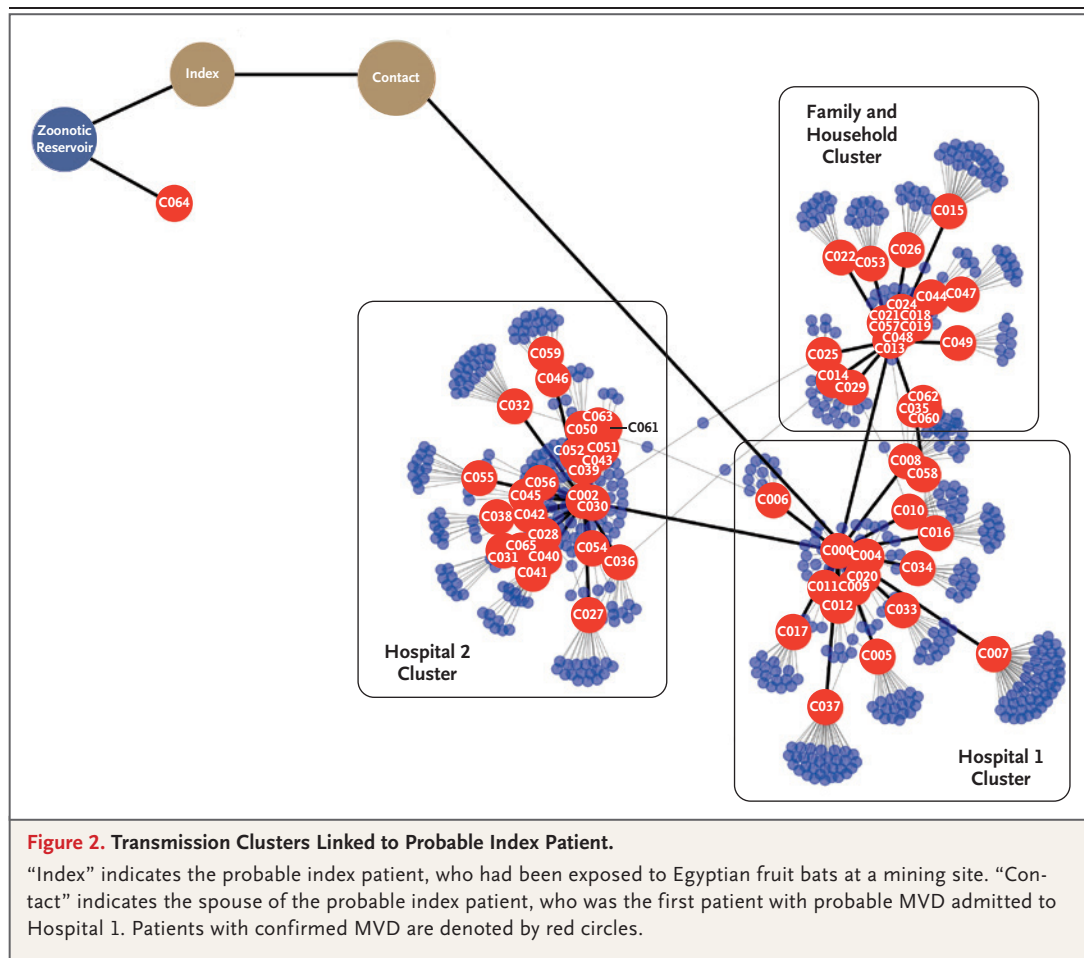
female. The median estimated incubation period (time from first known contact to onset of symptoms) was 10 days (interquartile range, 8 to 13), and symptom onset occurred a median of 2 days (interquartile range, 1 to 3) before hospital admission. Fifteen patients died, which represents a case fatality rate of 23%. Among the patients with confirmed MVD, 48% of those with a Ct value of less than 25 (indicating a high viral load) died, as compared with 5% of those with a Ct value of 25 or greater. Other demographic and clinical characteristics are shown in Table 2.

The most commonly reported symptoms at presentation included fever (in 95% of patients), fatigue (in 88%), and gastrointestinal symptoms such as nausea, vomiting, or diarrhea (in 82%). Hemorrhagic manifestations, such as epistaxis, gingival bleeding, or gastrointestinal hemorrhage, occurred in 42% of cases. Neurologic symptoms, such as confusion or altered mental status, were reported in 17% of cases. The full

spectrum of clinical symptoms is detailed in Table S2.

Baseline laboratory testing commonly showed elevations in hepatic enzyme levels, electrolyte disturbances, and thrombocytopenia, which typically worsened during the course of hospitalization. Approximately one third of patients had serum aspartate aminotransferase (AST) and alanine aminotransferase (ALT) levels greater than 3 times the upper limit of the normal range, and this degree of transaminitis developed during hospitalization in nearly 70% of patients. Hyponatremia (in 88% of patients), hypokalemia (in 39%), and hypoglycemia (in 30%) were the most frequently observed electrolyte disturbances. Additional laboratory findings are summarized in Table S3.

Supportive care included resuscitation with crystalloid fluids and close monitoring of volume status. Patients with hemorrhagic complications received transfusions of packed red cells,



**Table 2. Characteristics of Patients with Laboratory-Confirmed MVD According to Outcome.\***

| Characteristic                                                                          | Patients with<br>Confirmed MVD<br>(N = 66) | Outcome†         |                  |
|-----------------------------------------------------------------------------------------|--------------------------------------------|------------------|------------------|
|                                                                                         |                                            | Recovered        | Died             |
| Age — no./total no. (%)                                                                 |                                            |                  |                  |
| <35 yr                                                                                  | 40/66 (61)                                 | 31/40 (78)       | 9/40 (22)        |
| ≥35 yr                                                                                  | 26/66 (39)                                 | 20/26 (77)       | 6/26 (23)        |
| Sex — no./total no. (%)                                                                 |                                            |                  |                  |
| Female                                                                                  | 21/66 (32)                                 | 15/21 (71)       | 6/21 (29)        |
| Male                                                                                    | 45/66 (68)                                 | 36/45 (80)       | 9/45 (20)        |
| Duration of hospitalization — no./total no. (%)                                         |                                            |                  |                  |
| <7 days                                                                                 | 27/66 (41)                                 | 15/27 (56)       | 12/27 (44)       |
| ≥7 days                                                                                 | 39/66 (59)                                 | 36/39 (92)       | 3/39 (8)         |
| Patient occupation — no./total no. (%)                                                  |                                            |                  |                  |
| Health care worker                                                                      | 51/66 (77)                                 | 40/51 (78)       | 11/51 (22)       |
| Non-health care worker                                                                  | 15/66 (23)                                 | 11/15 (73)       | 4/15 (27)        |
| Province of residence — no./total no. (%)                                               |                                            |                  |                  |
| City of Kigali                                                                          | 57/66 (86)                                 | 45/57 (79)       | 12/57 (21)       |
| Other provinces                                                                         | 9/66 (14)                                  | 6/9 (67)         | 3/9 (33)         |
| Median initial Ct value on RT-PCR (IQR)                                                 | 25.8 (21.5–28.8)                           | 26.5 (21.5–28.7) | 19.4 (16.3–23.0) |
| Initial RT-PCR Ct value — no. (%)                                                       |                                            |                  |                  |
| <25                                                                                     | 27/66 (41)                                 | 14/27 (52)       | 13/27 (48)       |
| ≥25                                                                                     | 39/66 (59)                                 | 37/39 (95)       | 2/39 (5)         |
| Median time from first known contact to symptom onset (IQR) — days                      | 10 (8–13)                                  | 10 (7–12)        | 9 (7–11)         |
| Median time from symptom onset to hospital admission (IQR) — days                       | 2 (1–3)                                    | 1 (1–3)          | 3 (3–5)          |
| Median time from MVD diagnosis to first negative RT-PCR test (IQR) — days               | —                                          | 10.7 (7–13)      | —                |
| Median time from hospitalization to discharge among patients who recovered (IQR) — days | —                                          | 12 (9–15)        | —                |
| Median time from hospitalization to death among patients who died (IQR) — days          | —                                          | —                | 5 (3–6)          |

\* Ct denotes cycle threshold, IQR interquartile range, and RT-PCR reverse-transcriptase polymerase chain reaction.

† Among the 66 patients with confirmed MVD, 51 recovered and 15 died.

platelets, and fresh-frozen plasma. Empirical broad-spectrum antibiotic therapy and antifungal therapy were administered in cases of suspected secondary bacterial or fungal infections.

Clinical management of cases incorporated two investigational therapeutics: remdesivir, a nucleotide analogue prodrug that inhibits viral RNA polymerase, and MBP091, a monoclonal antibody targeting the Marburg virus glycoprotein. Given the absence of approved antiviral treatments for MVD, these therapies were ob-

tained through emergency procurement from international sources and deployed under an expanded-access protocol approved by the Rwanda National Ethics Committee and the Rwanda Food and Drugs Authority. Institutional review board approvals were expedited, which allowed remdesivir administration to begin on September 30 and MBP091 infusions to begin on October 5. Written informed consent was obtained from all patients who received an investigational treatment.

**Table 3. Complications and Clinical Interventions among Patients Hospitalized for MVD.**

| Complication or Intervention         | Patients Who Recovered<br>(N = 51) | Patients Who Died<br>(N = 15) |
|--------------------------------------|------------------------------------|-------------------------------|
| no. of patients (%)                  |                                    |                               |
| Complications                        |                                    |                               |
| Encephalopathy                       | 4 (8)                              | 11 (73)                       |
| Hypoxemia*                           | 7 (14)                             | 13 (87)                       |
| Hypercapnic respiratory failure      | 2 (4)                              | 3 (20)                        |
| Hypotension†                         | 6 (12)                             | 10 (67)                       |
| Acute kidney injury‡                 |                                    |                               |
| KDIGO stage 1                        | 7 (14)                             | 7 (47)                        |
| KDIGO stage 2                        | 2 (4)                              | 7 (47)                        |
| KDIGO stage 3                        | 1 (2)                              | 8 (53)                        |
| Coinfections                         |                                    |                               |
| Malaria test requested               | 25 (49)                            | 9 (60)                        |
| Malaria test reported positive       | 3 (6)                              | 1 (7)                         |
| Bloodstream infection                | 12 (24)                            | 4 (27)                        |
| Urinary tract infection              | 0                                  | 0                             |
| Hospital-acquired pneumonia          | 1 (2)                              | 1 (7)                         |
| Radiographic findings                |                                    |                               |
| Pulmonary infiltrate, lobar          | 1 (2)                              | 0                             |
| Pulmonary infiltrate, multifocal     | 2 (4)                              | 2 (13)                        |
| Pleural effusion                     | 2 (4)                              | 5 (33)                        |
| Supportive care interventions        |                                    |                               |
| Supplemental oxygen                  | 7 (14)                             | 13 (87)                       |
| Noninvasive ventilation              | 0                                  | 1 (7)                         |
| Endotracheal intubation              | 2 (4)                              | 7 (47)                        |
| Placement of central venous catheter | 3 (6)                              | 4 (27)                        |
| Intravenous fluid therapy            | 48 (94)                            | 13 (87)                       |
| Urinary catheterization              | 6 (12)                             | 12 (80)                       |
| Vasopressor therapy                  | 1 (2)                              | 4 (27)                        |
| Hemodialysis                         | 0                                  | 2 (13)                        |
| Antibiotic agent                     | 41 (80)                            | 11 (73)                       |
| Antimalarial agent                   | 3 (6)                              | 1 (7)                         |
| Packed red-cell transfusion          | 3 (6)                              | 3 (20)                        |
| Platelet transfusion                 | 3 (6)                              | 3 (20)                        |
| Fresh-frozen plasma transfusion      | 2 (4)                              | 3 (20)                        |
| Cryoprecipitate transfusion          | 1 (2)                              | 1 (7)                         |
| Disease-specific treatments          |                                    |                               |
| Remdesivir                           | 49 (96)                            | 3 (20)                        |
| MBP091                               | 8 (16)                             | 2 (13)                        |

\* Hypoxemia was defined as a blood oxygen saturation of less than 90%.

† Hypotension was defined as a systolic blood pressure of less than 90 mm Hg.

‡ Kidney Disease: Improving Global Outcomes (KDIGO) stages range from 1 to 3, with higher stages indicating more severe kidney injury.

Remdesivir was administered intravenously to 52 patients, including all critically ill persons who received supplemental oxygen or organ support. Rwanda also provided remdesivir as post-exposure prophylaxis for more than 150 health care workers. MBP091 was given to 10 patients: 6 received MBP091 under the expanded-access protocol, and 4 received it after enrolling in a randomized clinical trial coordinated by the WHO.<sup>16</sup> No immediate drug-related adverse events were identified with either therapy. A description of outcomes stratified according to clinical characteristics and treatment exposures is provided in Table 3.

Respiratory failure leading to tracheal intubation and mechanical ventilation developed in 9 patients, most often as a result of acute lung injury, refractory shock, or progressive metabolic acidosis. Two of the 9 patients were successfully extubated and discharged, and the remaining 7 died. Hemodialysis was initiated in 2 patients with severe acute kidney injury and multiorgan failure, both of whom died. A summary of clinical complications encountered during the treatment of hospitalized patients and interventions received is provided in Table S3.

#### VACCINATION IN THE OUTBREAK RESPONSE

Although no vaccines for MVD have been licensed, several candidates are in clinical development and have been prioritized by the WHO for emergency deployment during outbreaks. When the outbreak was declared, the government of Rwanda rapidly mobilized efforts to introduce the chimpanzee adenovirus 3–vectored Marburg virus vaccine (ChAd3-MARV) developed by the Sabin Vaccine Institute, which had been shown to be safe in a phase 1 trial.<sup>17</sup> In coordination with U.S. Biomedical Advanced Research and Development Authority (BARDA) and the Coalition for Epidemic Preparedness Innovations (CEPI), 700 doses were shipped to Rwanda on October 5, 2024. Concurrently, the Rwanda Food and Drugs Authority and the Rwanda National Ethics Committee expedited regulatory approvals, authorizing emergency use on October 2 and October 7, respectively. The first frontline health care worker provided informed consent and received the vaccine on October 10, just 13 days after the outbreak was declared.

By the end of October 2024, a total of 1710 frontline workers and high-risk contacts had

been vaccinated, with administration prioritized for health care workers, burial teams, and laboratory personnel. In addition to emergency deployment, Rwanda initiated a phase 2 clinical trial to assess the safety and immunogenicity of the vaccine under a modified WHO Solidarity core protocol, adapting the design to optimize protection of critical health care personnel.<sup>18</sup> The first patient was enrolled on October 15, and no serious vaccine-related adverse events were reported during the emergency rollout.

#### OUTBREAK CONTAINMENT

By the third week of the response, after the rapid implementation of multisectoral, comprehensive response activities, the outbreak trajectory shifted (Fig. S1). An outpatient clinic dedicated to monitoring and psychosocial support for recovered patients opened on October 23. The final patient was discharged from the treatment center on November 11. On December 20, no new confirmed cases had been reported for more than two incubation periods, which signaled that transmission had been successfully interrupted, and the outbreak was declared over.

#### DISCUSSION

This MVD outbreak in Rwanda represents the first documented occurrence of the virus in the country and emergence in a densely populated urban setting. Unlike past MVD outbreaks, which have been marked by high case fatality and delayed detection in remote regions, this outbreak saw the rapid initiation of a multisectoral response, which is likely to have contributed to a lower case fatality (23%). Although direct comparisons across outbreaks are limited by differences in surveillance, health care capacity, and diagnostic access, this experience underscores the potential effect of early case identification, aggressive supportive care, and access to investigational treatments in mitigating mortality from MVD.

In this outbreak, nosocomial transmission accounted for 77% of cases, which underscores the need to strengthen infection prevention and control measures in health care settings in areas where viral hemorrhagic fever is endemic, especially in the context of resuscitation of critically ill patients. An important challenge in this setting was that many of the affected health care

workers were caring for critically ill colleagues, often under stressful conditions.<sup>19,20</sup> Studies of previous filovirus disease outbreaks suggest that familiarity with infected patients, high emotional burden, and emergency clinical interventions may contribute to lapses in infection prevention and control measures.<sup>21</sup> Improvements in the capacity for testing and earlier consideration of MVD in the differential diagnosis of severe febrile illness are also necessary, particularly in high-risk regions where viral hemorrhagic fever may initially be unrecognized and mimic other endemic febrile illnesses, such as malaria.

Although access to advanced supportive care contributed to survival in this cohort, investigational treatments may have also played a role. Formal efficacy assessment of these interventions is not possible outside a clinical trial. Among the 52 patients treated with remdesivir, 3 (6%) died, as compared with 12 of the 14 patients (86%) who did not receive it. Among the 10 patients treated with MBP091, 2 (10%) died, as compared with 13 of the 56 patients (23%) who did not receive it. Because neither treatment was given in a randomized fashion, other factors may be contributing to the relatively low mortality observed. This outbreak further showed that organ-support interventions, including dialysis and vasopressor therapy, can be administered safely for filovirus infections with proper infection prevention and control measures in place, and these interventions should be standard care in future outbreaks.

The deployment of the ChAd3-MARV vaccine within 13 days after declaration of the outbreak, alongside a concurrent phase 2 clinical trial, shows that investigational vaccines can be integrated into an outbreak response. Although the effect of the vaccine on transmission was not assessed directly, the prioritization of frontline workers for emergency vaccination under an adapted WHO Solidarity core protocol provides an important model for responses to future filovirus disease outbreaks.

The multisectoral public health investigation and response during the Rwanda MVD outbreak highlights the importance of an integrated and multidisciplinary One Health approach to strengthening future pandemic preparedness, prevention, and response activities.<sup>22</sup> The integration of surveillance systems and support from genomic analysis can strengthen the capacity for earlier

detection of MVD in humans, animals, and the environment.<sup>13</sup>

The prevention of high mortality in future filovirus disease outbreaks will require faster recognition of viral hemorrhagic fevers through a One Health approach involving collaboration among community health workers, human health and genomics specialists, bat biologists, ecologists, conservationists, and other relevant experts. Such an effort will strengthen infection prevention and control measures, reduce the risk of nosocomial transmission, and ensure timely delivery of supportive and intensive care, as well as psychosocial support for patients, families, and staff. Furthermore, sustainable investments in clinical trial infrastructure and vaccine stockpiling will be essential to ensuring that investigational countermeasures can be deployed swiftly when outbreaks emerge. These efforts will be critical to reducing morbidity and mortality from future filovirus disease outbreaks and strengthening global health security against emerging pathogens.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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